ANALYTICS PORTFOLIO PROJECT

Brazilian E-Commerce Analytics Dashboard

End-to-end business intelligence project analyzing sales performance, logistics efficiency, and customer satisfaction using the Olist marketplace dataset.

R$15.84M
TOTAL REVENUE

99K
ORDERS ANALYZED

12.50d
AVG. DELIVERY TIME

4.09 / 5
AVG. REVIEW SCORE

Power BI · Power Query · DAX · Snowflake Schema · Data Modeling · Business Analytics

Dataset · Olist Brazilian E-Commerce (Kaggle) · 2016–2018

Scope · Sales · Product Performance · Logistics · Customer Satisfaction

01 EXECUTIVE SUMMARY

At a Glance

This project delivers a multi-page Power BI dashboard built on the Olist Brazilian E-Commerce dataset (~100,000 orders, 2016–2018). The analysis spans four business dimensions: revenue performance, product category profitability, logistics efficiency, and customer satisfaction. A Snowflake schema was implemented to handle the dataset's multiple fact tables, and DAX measures were engineered to surface actionable KPIs. The central finding — that longer delivery times correlate negatively with customer review scores — points directly to logistics optimization as the highest-leverage business lever.

02 PROJECT OVERVIEW

Objective & Scope

What this project set out to answer and how.

Olist is Brazil's largest online retail marketplace aggregator. The dataset captures the full order lifecycle — from purchase to delivery and customer feedback — across multiple product categories and thousands of sellers nationwide.

The objective was to build a business intelligence dashboard that enables stakeholders to monitor commercial performance, identify top-performing categories, track logistics efficiency, and understand the relationship between operational quality and customer satisfaction.

→ PRIMARY QUESTIONS

- → Which product categories drive the most revenue?
- → How does delivery time affect customer satisfaction?
- → How has monthly revenue trended over time?
- → What percentage of freight cost is attributable to revenue?

R$15.84M	99K	12.50	4.09	R$2.25M
TOTAL REVENUE	TOTAL ORDERS	AVG. DELIVERY DAYS	AVG. REVIEW SCORE	TOTAL FREIGHT

03 DATASET & DATA PREPARATION

Source, Scope & Cleaning

Olist Brazilian E-Commerce Public Dataset — Kaggle · ~100,000 orders · 2016–2018

TABLE	DESCRIPTION	ROLE
orders	Order IDs, statuses, dates, delivery timestamps	Central fact hub
order_items	Product-level line items, prices, freight values	Fact (revenue)
order_payments	Payment type, installments, value	Fact (payments)
order_reviews	Review scores, review dates	Fact (satisfaction)
products	Category, physical dimensions	Dimension
customers	Customer location, unique IDs	Dimension
sellers	Seller location, seller IDs	Dimension

⚠ REVENUE ALIGNMENT NOTE

A key preparation decision involved filtering order statuses to **delivered**, **approved**, and **shipped** when aggregating total payments into the revenue KPI. This prevents inflated figures caused by cancelled or otherwise unresolved orders that still have payment records in the dataset.

STEP 1

Table Rename & Column Pruning

Renamed all tables for readability and removed columns with no analytical value to reduce model size.

STEP 2

Primary Key Validation

Verified uniqueness of dimension table primary keys (customer_id, product_id, seller_id) to ensure clean one-to-many join integrity.

STEP 3

Date Formatting

Standardised all date columns and created a dynamic Calendar table to support time-intelligence functions and monthly slicing.

STEP 4

Status Filtering for Revenue

Applied logical order-status filters to align payment values with actual fulfilled revenue, eliminating noise from cancelled orders.

Snowflake Schema Design

Why a Snowflake schema was chosen over a Star schema, and how filter propagation was handled.

▶ SCHEMA DECISION

The dataset contains **three separate fact tables**: order_items, order_payments, and order_reviews. A traditional Star schema would create many-to-many relationship conflicts between these facts and shared dimensions, causing incorrect aggregations and slicer failures. A **Snowflake schema** resolves this by normalising the fact tables through the central orders table, preserving data integrity throughout.

▶ CROSS-FILTER DIRECTION

Default single-direction filtering prevents slicers from propagating across fact tables. Cross-filter direction was set to **"Both"** on:

`order_items ↔ orders`

`order_reviews ↔ orders`

This ensures that the Product Category and Review Score slicers correctly filter all visuals simultaneously.

FACT TABLES

`order_items` `order_payments` `order_reviews`

DIMENSION TABLES

`orders (hub)` `products` `customers` `sellers` `Calendar dynamic`

Relationships: Many-to-one (facts → orders), One-to-many (orders → dimensions) | Cross-filter: Both on order_items ↔ orders and order_reviews ↔ orders

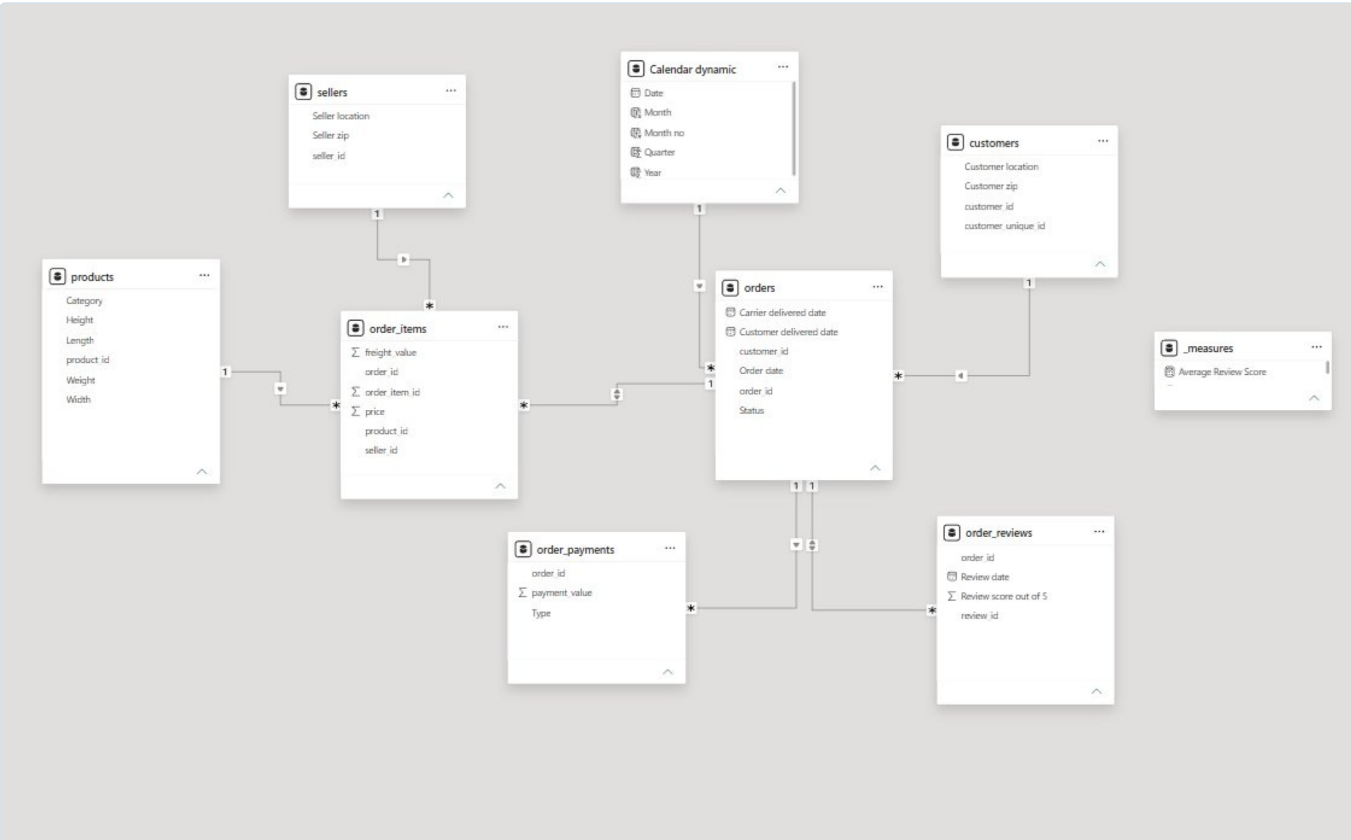

Fig. 4 — Snowflake schema in Power BI model view. The orders table acts as the central hub connecting all three fact tables to their shared dimensions.

DAX Measures

`DISTINCTCOUNT(order_id)` `SUMX(order_items, price)` `AVGX(orders, DATEDIFF(...))` `AVERAGEX(order_reviews, score)` `SUMX(order_items, freight_value)`

All KPI measures were built as explicit DAX measures in a dedicated `_measures` table, separating business logic from raw data tables for maintainability and reuse.

Visualizations & Interactivity

Four visuals — each answering a distinct business question.

Fig. 1 — Main dashboard. Interactive slicers (Date, Product Category, Order Status, Review Score) filter all visuals simultaneously via the Snowflake schema.

Monthly Revenue Trend — A line chart showing monthly revenue from September 2016 through October 2018. Revenue grew steadily through 2017, peaked at ~R$1.58M in November 2017 (consistent with Black Friday season), then trended toward a post-holiday normalization.

Top Product Categories by Revenue — A ranked table of product categories with total sales. Health & Beauty leads at R$1.26M, followed by Bed/Bath/Table at R$1.04M. Categories like Automotive and Christmas Articles trail significantly, signaling clear portfolio concentration.

06 KEY INSIGHTS & ANALYSIS

What the Data Reveals

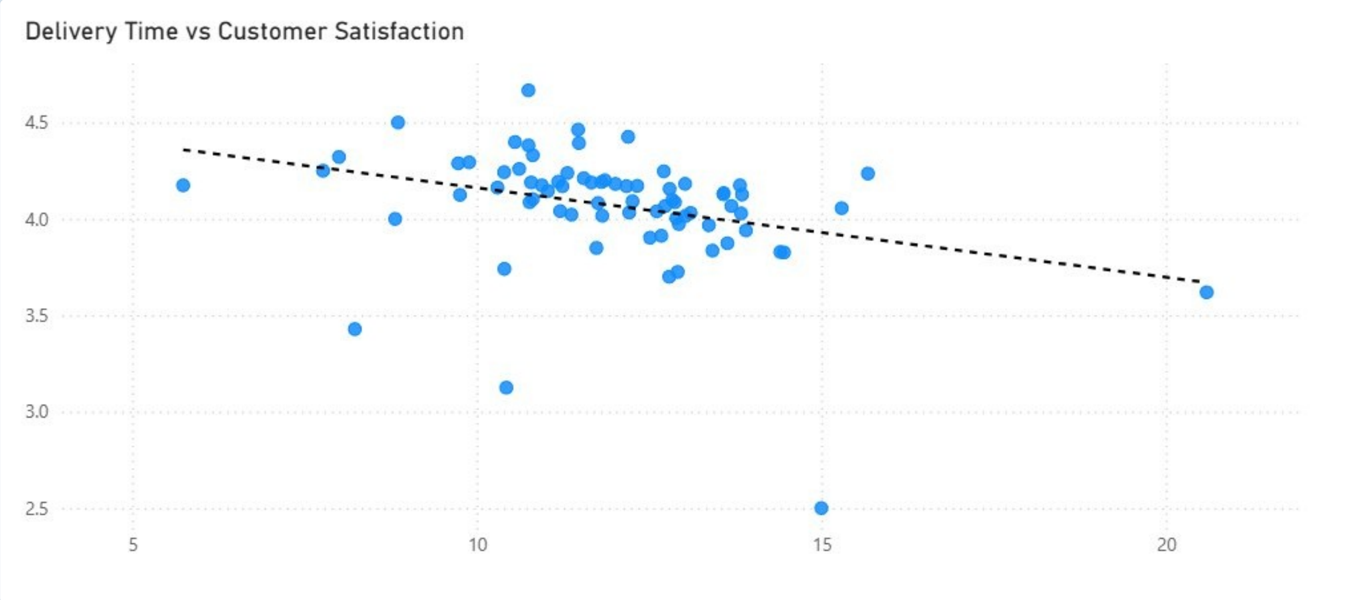

Fig. 2 — Scatter plot: Delivery Time (days) vs. Average Review Score. Each point represents an aggregated delivery cohort. The negative trendline is statistically clear.

↘ CORE FINDING — LOGISTICS DRIVES SATISFACTION

The scatter plot reveals a consistent **negative relationship** between delivery duration and customer satisfaction. Orders delivered in 6–8 days cluster around review scores of 4.2–4.5, while those exceeding 14 days drop toward 3.6–3.8. The trendline slope suggests that each additional week of delivery time is associated with approximately a **0.3–0.4 point decline** in review score. For a marketplace business, this is a direct link between operational efficiency and revenue-generating reputation.

INSIGHT	FINDING	IMPLICATION
Revenue Seasonality	Peak at R$1.58M in Nov 2017	Align inventory and logistics ahead of Nov–Dec season
Category Concentration	Top 3 categories = ~40% of revenue	Risk exposure to narrow category mix; diversification needed
Delivery vs. Review	Negative correlation (r ≈ −0.5)	Cutting avg. delivery time by ~2 days could lift review scores materially
Freight as % of Revenue	R$2.25M / R$15.84M ≈ 14%	Freight optimization has substantial P&L impact
Review Score	4.09 average on 5-point scale	Solid baseline; further gains achievable via logistics improvements

Olist E-Commerce Analytics Dashboard

Summary

olist

Data Modeling and Preparation:

- Renamed tables for clarity and removed unnecessary columns.
- Ensured uniqueness of primary keys in dimension tables.
- Applied filters and calculated measures (Total Orders, Total Revenue, Total Freight, Avg Delivery Days) using DAX functions (DISTINCTCOUNT, SUMX, AVGX + DATEDIFF).
- Adjusted date formats for dynamic visualizations.

Schema and Relationships:

- Implemented a Snowflake schema with fact tables: order_items, order_payments, order_reviews.
- Used many-to-one relationships for facts and one-to-many for dimensions.
- Set cross-filter direction to "Both" between order_items → orders and order_reviews → orders to ensure slicers (product category, review score) propagate correctly across all visuals.

Insights and Analytics:

- Applied logical filters on order status (delivered, approved, shipped) to align total payments with total revenue.
- Visualizations include:
  - -Monthly Revenue Trend
  - -Top Product Categories by Revenue
  - -Review Score vs Avg Delivery Days
  - -Delivery Time vs Customer Satisfaction (scatter plot with trendline)

Key Takeaways

- Dashboard shows operational performance, customer satisfaction, and category-level insights.
- Demonstrates ability to handle complex data models, DAX calculations, and meaningful business analysis.

Fig. 3 — Dashboard summary page documenting data modeling decisions, schema design, and key analytical takeaways.

07 BUSINESS RECOMMENDATIONS

Actionable Next Steps

Based on findings from the dashboard analysis.

▶ Invest in Last-Mile Logistics Optimization

The scatter plot data suggests that reducing average delivery time from 12.5 to 10 days could push average review scores above 4.2. Partnering with regional carriers in high-volume states would have the most direct impact.

▶ Double Down on Top-Revenue Categories

Health & Beauty and Bed/Bath/Table collectively drive outsized revenue. Expanding seller depth in these categories — particularly ahead of the November seasonality peak — would yield the highest marginal revenue gain.

▶ Audit Freight Cost Efficiency

At approximately 14% of gross revenue, freight costs represent a meaningful line item. A cost-per-category freight analysis could identify products where the freight-to-price ratio is unfavorable and suggest pricing or fulfillment adjustments.

▶ Build a Seller Performance Scorecard

The dataset contains sufficient seller-level data to build a secondary view scoring sellers on delivery speed, review outcomes, and order volume — enabling data-driven seller support and incentive programmes.

08 SKILLS DEMONSTRATED

Technical Competencies

⬡ Power BI & Power Query

End-to-end dashboard development, data transformation, calculated columns, and dynamic calendar tables.

⬡ DAX Engineering

DISTINCTCOUNT, SUMX, AVERAGEX, DATEDIFF — purpose-built KPI measures separated into a dedicated measures table.

⬡ Data Modeling

Snowflake schema design with multiple fact tables, managed cross-filter direction, and relationship integrity.

⬡ Business Analytics

Translating raw e-commerce data into revenue, logistics, and satisfaction insights with clear business narratives.

⬡ Data Preparation

Multi-table cleaning, primary key validation, status-based filtering for revenue accuracy, and date standardisation.

⬡ Data Visualization

KPI cards, trend lines, scatter plots with trendlines, ranked tables, and interactive cross-filtering slicers.

Conclusion

This project demonstrates a full-stack business analytics workflow — from raw data ingestion and cleaning through data modeling, DAX engineering, and insight generation — resulting in a production-grade Power BI dashboard. The central business finding is clear and actionable: **logistics performance is the single most controllable driver of customer satisfaction** in the Olist marketplace. Reducing average delivery time by even two days would likely yield a measurable improvement in review scores, with downstream effects on platform reputation, repeat purchase rates, and seller retention.